

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CRYPTOGRAPHY AND NETWORK SECURITY

COURSE CODE: CSA5110

ASSIGNMENT REPORT

NAME	NOEL RAJU
REGISTER NUMBER	192372315
COURSE	CRYPTOGRAPHY AND NETWORK SECURITY
ASSESSMENT	ASSESSMENT TOOL 2

Submitted as part of the course assessment

1. INTRODUCTION

Cryptography is the science of protecting information by transforming readable data, called plaintext, into an unreadable form called ciphertext. In modern computer networks, cryptography is essential because financial records, passwords, customer information, banking transactions, and business data frequently travel through communication channels that may be observed or attacked by unauthorized parties.

Block ciphers are an important class of symmetric-key encryption algorithms. A block cipher processes plaintext in fixed-size blocks using the same secret key for encryption and decryption. However, a secure block cipher alone is not enough. The mode of operation determines how individual blocks are combined and therefore has a major effect on confidentiality, pattern hiding, error behavior, and practical security.

This assessment studies two practical cryptographic decisions. The first problem examines why Electronic Codebook (ECB) mode reveals patterns when a company repeatedly transmits similar financial data, and it identifies a safer mode for that environment. The second problem compares DES, 3-DES, and Blowfish for a banking application that requires strong security with moderate performance.

Objectives of the Assessment

- Explain the operation and weakness of ECB mode when plaintext blocks repeat.
- Recommend a safer block cipher mode for repeated financial data transmission.
- Compare DES, 3-DES, and Blowfish in terms of key strength, practical security, and efficiency.
- Select the most appropriate algorithm among the given choices for a banking system.

Basic Security Requirements

A financial or banking system normally requires confidentiality, integrity, authentication, and availability. Encryption mainly supports confidentiality, but the selected mode should also avoid leaking structural information. In real deployments, authenticated encryption is preferred because it can protect both confidentiality and integrity.

2. SECURE DATA TRANSMISSION USING BLOCK CIPHER

2.1 Scenario

A company transmits repeated financial data blocks over a network. It initially uses ECB mode and notices visible patterns in the ciphertext. This is a serious warning because encrypted data should not reveal obvious relationships between repeated portions of the original information.

2.2 How ECB Mode Works

Electronic Codebook mode divides a message into fixed-size blocks and encrypts each block independently with the same secret key. If the plaintext blocks are represented as P_1 , P_2 , P_3 and so on, the ciphertext blocks are obtained independently as $C_1 = E(K, P_1)$, $C_2 = E(K, P_2)$, $C_3 = E(K, P_3)$, where E is the block-cipher encryption function and K is the secret key.

The main advantage of ECB is simplicity. Blocks can be encrypted or decrypted independently, and processing can be parallelized. However, this independence creates its major security weakness: there is no randomization or chaining relationship between blocks.

2.3 Answer to Question 1(A): Why Patterns Occur in ECB

Patterns occur because ECB is deterministic for a fixed key. Whenever two plaintext blocks are identical, their ciphertext blocks are also identical. Therefore, repeated account fields, transaction formats, fixed headers, repeated zero values, repeated amounts, or standard financial record structures can create repeated ciphertext blocks.

An observer may not immediately know the plaintext values, but repeated ciphertext still reveals equality and structure. This is called information leakage. For example, if a company repeatedly sends a record containing the same department code or a standard payment status, an attacker can notice that the corresponding encrypted block appears again and again.

3. SECURITY IMPACT OF ECB PATTERN LEAKAGE

3.1 Loss of Semantic Security

Good encryption should make ciphertext appear unrelated to the structure of the plaintext. ECB fails to provide this property for repeated blocks. Two identical messages or identical sections of a message can generate recognizable repetitions, allowing traffic analysis even without knowledge of the encryption key.

3.2 Example with Financial Data

Assume a company sends fixed-length records such as Account Type, Branch Code, Transaction Category, Amount Range, and Status. If some of these fields repeat and align with block boundaries, ECB produces the same encrypted block for each repeated value. An attacker monitoring the network could build a frequency profile and correlate recurring ciphertext blocks with recurring business events.

Plaintext Block	ECB Result	Security Observation
PAYMENT-APPROVED	Cipher Block X	Same result every time
PAYMENT-APPROVED	Cipher Block X	Pattern is visible
PAYMENT-PENDING	Cipher Block Y	Different block
PAYMENT-APPROVED	Cipher Block X	Repetition can be tracked

3.3 Additional ECB Weaknesses

- Identical plaintext blocks produce identical ciphertext blocks.
- Block positions and repeated structures can be inferred.
- An attacker may rearrange independent ciphertext blocks in some protocols if integrity protection is absent.
- ECB does not use an initialization vector or nonce to randomize repeated plaintext.

3.4 Conclusion for Part A

The company sees patterns because ECB encrypts each block independently and deterministically. For repeated financial records, this behavior is unsuitable because structural information is valuable to an attacker even when the actual secret key remains unknown.

4. ANSWER TO QUESTION 1(B): RECOMMENDED BLOCK CIPHER MODE

4.1 Recommended Choice: AES-GCM

For a modern networked financial-data scenario, AES in Galois/Counter Mode (AES-GCM) is a strong choice. GCM is an authenticated-encryption mode. It uses counter-based encryption so that repeated plaintext blocks do not simply produce repeated ciphertext blocks when a unique nonce is used correctly. It also produces an authentication tag that allows the receiver to detect unauthorized modification.

The question asks for a more secure block cipher mode. A traditional answer could be CBC because chaining and a random initialization vector hide repeated-block patterns. However, for a real company transmitting financial information over a network, GCM is preferable because it provides both confidentiality and integrity in one widely used construction.

4.2 Why GCM Fits the Scenario

- Pattern hiding: counter values cause encryption inputs to differ from block to block.
- Integrity protection: the authentication tag detects changes to ciphertext and authenticated metadata.
- High performance: counter-mode encryption can be parallelized and is efficient in hardware and software.
- Network suitability: GCM is widely used in modern secure communication protocols.
- No padding requirement for arbitrary-length messages, unlike classic CBC constructions.

4.3 Critical Nonce Requirement

GCM requires a unique nonce for every encryption performed with the same key. Reusing a nonce with the same key can seriously damage confidentiality and authentication security. Therefore, the company must implement reliable nonce generation or management, key rotation, and authenticated-tag verification.

4.4 If CBC Is Required by the Syllabus

Cipher Block Chaining (CBC) is also more secure than ECB for hiding repeated patterns. In CBC, each plaintext block is XORed with the previous ciphertext block before encryption, while the first block uses an unpredictable initialization vector. As a result, equal plaintext blocks at different positions normally encrypt differently. However, CBC by itself does not provide integrity, so it should be combined with a secure message authentication mechanism.

5. ECB, CBC AND GCM COMPARISON

Feature	ECB	CBC	GCM
Pattern hiding	Poor	Good with proper IV	Excellent with unique nonce
Block dependency	Independent	Chained	Counter based
Integrity included	No	No	Yes, authentication tag
Parallel encryption	Yes	Limited	Yes
Padding	Usually required	Usually required	Not required in the same way
Error behavior	Local block effect	May affect current/next block	Tampering rejected when tag checked
Suitability here	Not recommended	Better than ECB	Recommended

5.1 Proposed Transmission Process

The sender should use a strong secret key, create a unique nonce for each message, encrypt the financial plaintext using AES-GCM, and transmit the nonce, ciphertext, and authentication tag. The receiver uses the same secret key and the received nonce to verify the tag before accepting the decrypted data.

5.2 Security Result

This design prevents the simple repeated-block leakage seen in ECB and also helps detect active manipulation. Therefore, changing only from ECB to a randomized or nonce-based secure mode greatly improves the confidentiality of repeated financial data. Using authenticated encryption further strengthens the design by preventing modified ciphertext from being silently accepted.

6. CHOOSING AN ENCRYPTION ALGORITHM FOR A BANKING SYSTEM

6.1 Scenario and Requirements

A banking application requires high security and moderate performance for encrypting customer transactions. The available choices are DES, 3-DES, and Blowfish. Because the options are older symmetric ciphers, the selection must balance practical security strength against computational cost.

6.2 DES

Data Encryption Standard (DES) is a 64-bit block cipher with an effective key length of 56 bits. It was historically important, but a 56-bit key is far too small for modern high-security applications. Exhaustive key search against DES is practical with specialized resources, so DES should not be selected for protecting current banking transactions.

6.3 Triple DES (3-DES)

Triple DES applies the DES operation multiple times, commonly using a keying arrangement that gives substantially more security than single DES. It was used for many years in financial systems because it allowed organizations to extend the life of existing DES-based infrastructure. Its main disadvantage is speed: performing DES three times makes it considerably slower than newer ciphers.

3-DES also retains a 64-bit block size, which limits the amount of data that should be safely processed under a single key in modern high-volume environments. For new systems it is generally considered a legacy choice rather than a preferred modern design.

6.4 Blowfish

Blowfish is a symmetric 64-bit block cipher designed with a variable key length of 32 to 448 bits. It is generally faster in software than 3-DES and, when used with a sufficiently strong key, offers much greater brute-force resistance than DES. Its key setup is relatively expensive, but once subkeys are prepared, bulk encryption can be efficient.

7. ANSWER TO QUESTION 2(A): MOST SUITABLE ALGORITHM

7.1 Selected Algorithm: Blowfish

Among the three choices given in the assessment—DES, 3-DES, and Blowfish—Blowfish is the most suitable when the stated goal is high security with moderate performance. DES is eliminated because its 56-bit key is insecure. 3-DES improves security but has a significant performance cost because of repeated DES operations. Blowfish supports much longer keys and is efficient in software after key setup.

7.2 Important Practical Qualification

This selection is specifically among the three algorithms listed in the question. For a new real-world banking system, modern standards such as AES, normally used in an authenticated mode such as GCM, would be preferred over all three choices. Blowfish has a 64-bit block size and is therefore not the ideal cipher for large modern transaction volumes. Nevertheless, within the restricted options, it gives the best overall balance requested by the scenario.

7.3 Why DES Is Rejected

- Only 56 effective key bits.
- Vulnerable to exhaustive brute-force search by modern standards.
- Does not satisfy the requirement for high security.

7.4 Why 3-DES Is Not the First Choice

- Stronger than DES, but computationally slow.
- Requires multiple DES operations for every block.
- Uses a small 64-bit block size and is considered legacy for new applications.
- Moderate performance is harder to achieve in high-volume banking workloads.

7.5 Why Blowfish Is Chosen

- Supports a large variable key length up to 448 bits.
- Offers better software efficiency than 3-DES in many conventional implementations.
- Provides a stronger security margin than DES against brute-force key search.
- Matches the assignment's combination of security strength and moderate performance better than the other listed options.

8. ANSWER TO QUESTION 2(B): SECURITY AND EFFICIENCY

JUSTIFICATION

Factor	DES	3-DES	Blowfish
Block size	64 bits	64 bits	64 bits
Key strength	56-bit effective key	Much stronger than DES	32-448-bit variable key
Brute-force resistance	Poor today	High relative to DES	High with strong key
Speed	Fast but insecure	Slow	Moderate/Fast after setup
Modern new-system status	Obsolete	Legacy	Legacy but stronger option here
Assessment suitability	Unsuitable	Possible but inefficient	Best among listed choices

8.1 Security Strength

The primary security problem with DES is the small key space. 3-DES increases resistance by applying DES repeatedly with multiple keys, but this security improvement comes with a heavy computational cost. Blowfish allows a much larger key and has no practical brute-force weakness comparable to 56-bit DES when a strong key is chosen.

8.2 Efficiency

For banking software, encryption may be performed for many transaction records. 3-DES requires repeated DES calculations, increasing CPU cost and latency. Blowfish performs an expensive key-expansion phase, but this cost can be acceptable when the same session or application key is used to process multiple blocks. Its data-encryption phase is comparatively efficient.

8.3 Overall Decision

Therefore, Blowfish is the best answer among the provided algorithms for a system that explicitly needs both strong security and moderate performance. The decision should still be accompanied by secure key generation, key storage, regular key rotation, a safe mode of operation, and integrity protection.

9. FINAL CONCLUSION AND RECOMMENDATIONS

The first scenario demonstrates that choosing a secure encryption algorithm is not sufficient unless it is used with a suitable mode of operation. ECB exposes patterns because identical plaintext blocks encrypted with the same key become identical ciphertext blocks. This makes ECB inappropriate for repeated financial records, where message structures and field values can recur frequently.

For secure data transmission, AES-GCM is recommended because it hides repeated patterns through nonce-based counter encryption and also provides an authentication tag for integrity. If the academic comparison is limited to traditional confidentiality modes, CBC with an unpredictable IV is clearly safer than ECB for pattern hiding, although CBC should be combined with authentication.

In the second scenario, DES, 3-DES, and Blowfish were compared. DES is unsuitable because its 56-bit effective key is too weak. 3-DES is stronger but slow and legacy. Blowfish provides the best balance among the listed options because it supports much larger keys and has better software performance than 3-DES in many settings.

Key Recommendations

- Do not use ECB for repeated or structured confidential information.
- Prefer authenticated encryption such as AES-GCM for modern network communication.
- Never reuse a GCM nonce with the same key.
- Among DES, 3-DES, and Blowfish, select Blowfish for this assessment scenario.
- For an actual new banking system, prefer modern standardized encryption such as AES rather than deploying DES, 3-DES, or Blowfish.

References

1. **National Institute of Standards and Technology (NIST)**, Recommendation for Block Cipher Modes of Operation: Galois/Counter Mode (GCM) and GMAC.
2. **National Institute of Standards and Technology (NIST)**, Data Encryption Standard and Triple Data Encryption Algorithm historical standards and guidance.
3. **Bruce Schneier**, Description and design of the Blowfish symmetric block cipher.
4. **William Stallings**, Cryptography and Network Security: Principles and Practice.